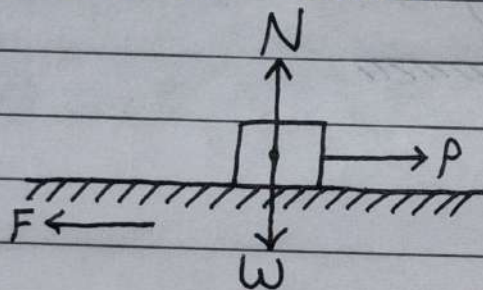


MODULE-II

Friction

Friction

When a body move or tends to move over another body, a force opposing the motion developed at the contact surface. The force that oppose the motion or to the tendency of motion is called frictional force.



Types of Friction:

1. Static Friction (μ_s):

The frictional force developed till the body starts moving is called static friction.

2. Dynamic Friction (μ_k):

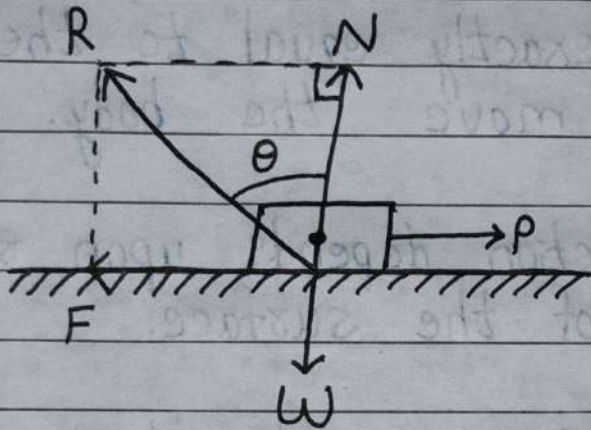
The frictional resistance developed at contact surface, when the body moves over another body is known as dynamic friction.

Coefficient of Friction (μ):

It is defined as the ratio of limiting friction to the normal reaction.

$$\mu = \frac{\text{Limiting friction}}{\text{Normal reaction}}$$

$$\mu = \frac{F}{N}$$



Angle of Friction (θ):

It is the angle between the normal reaction (N) and the resultant force (R).

$$\theta = \tan^{-1}\left(\frac{F}{N}\right)$$

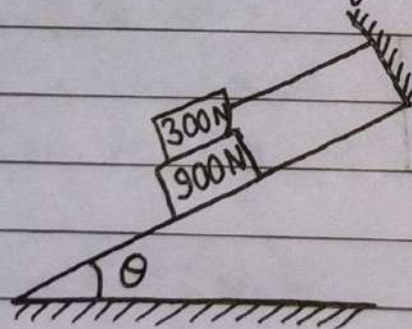
$$\Rightarrow \theta = \tan^{-1}(\mu)$$

$$\Rightarrow \mu = \tan \theta$$

Imp. # Laws of Coulomb Friction

- ① The force of friction always acts in the direction opposite to that in which the body moves or tends to move.
- ② Till the limiting value is reached, the magnitude of friction is exactly equal to the force which tends to move the body.
- ③ The force of friction depends upon roughness and smoothness of the surface.
- ④ The force of friction is independent of the area of contact.

Q) Find θ in the given figure?



$$\mu = \frac{1}{3}$$

Solⁿ:

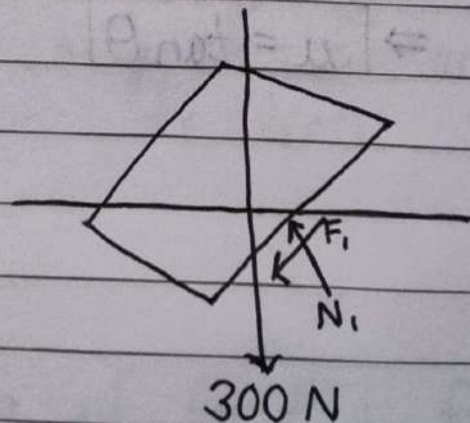
For 300 N Block,

$$N_1 = 300 \cos \theta$$

Also,

$$F_1 = \mu N_1 = \frac{1}{3} \times 300 \cos \theta$$

$$\Rightarrow F_1 = 100 \cos \theta$$

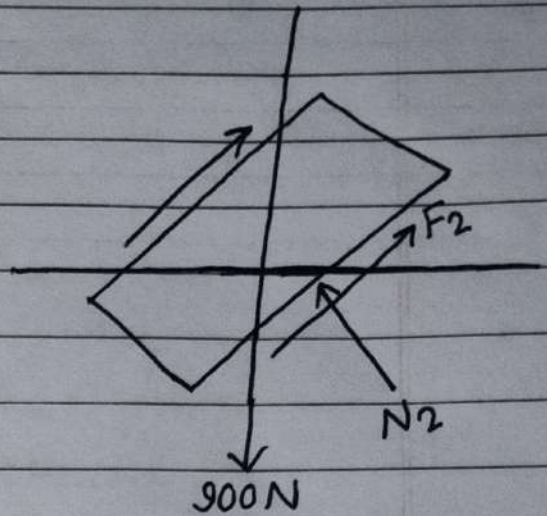


For 900 N Block,

$$N_2 = N_1 + 900 \cos \theta$$

$$\Rightarrow N_2 = 300 \cos \theta + 900 \cos \theta$$

$$\Rightarrow N_2 = 1200 \cos \theta$$



Also,

$$F_2 = \mu N_2 = \frac{1}{3} \times 1200 \cos \theta$$

$$\Rightarrow F_2 = 400 \cos \theta$$

At Equilibrium,

$$F_1 + F_2 = 900 \sin \theta$$

$$\Rightarrow 100 \cos \theta + 400 \cos \theta = 900 \sin \theta$$

$$\Rightarrow 500 \cos \theta = 900 \sin \theta$$

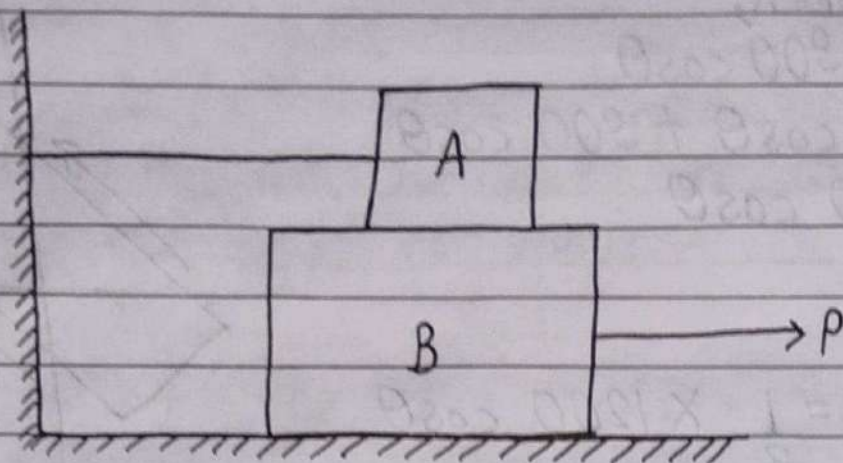
$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{500}{900}$$

$$\Rightarrow \tan \theta = \frac{5}{9}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{5}{9} \right)$$

$$\therefore \theta = 29.054^\circ$$

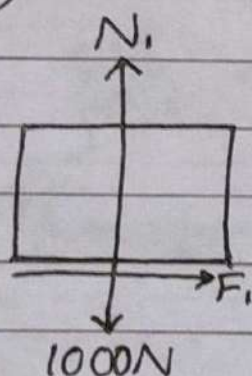
Q)



Block A weight 1000 N rest over block B which weights 2000 N as shown in figure. Block A is tied to a wall with a horizontal string. If the coefficient between A & B is $\frac{1}{4}$ and that between B & floor is $\frac{1}{3}$, what value of P is required to create impending motion if P is horizontal?

Solⁿ:

For Block A,

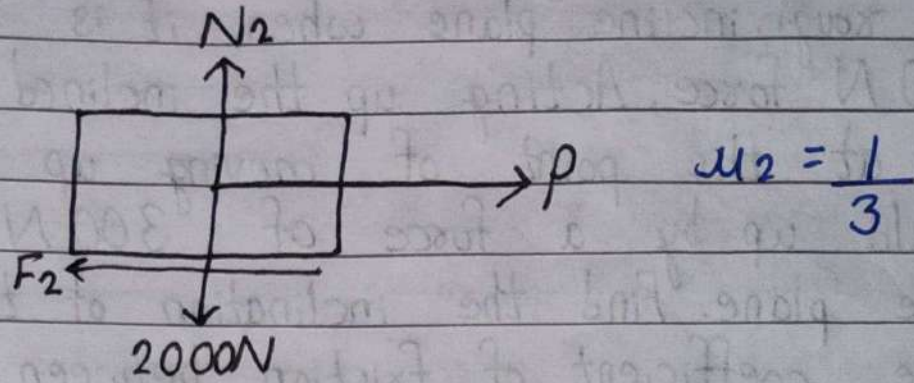


$$\mu_1 = \frac{1}{4}$$

$$N_1 = 1000\text{ N}$$

$$F_1 = \mu_1 \times N_1 = \frac{1}{4} \times 1000 = 250\text{ N}$$

For Block B,



$$N_2 = (N_1 + 2000) \text{ N} = (1000 + 2000) \text{ N}$$

$$\Rightarrow N_2 = 3000 \text{ N}$$

$$F_2 = \mu_2 \times N_2 = \frac{1}{3} \times 3000$$

$$\Rightarrow F_2 = 1000 \text{ N}$$

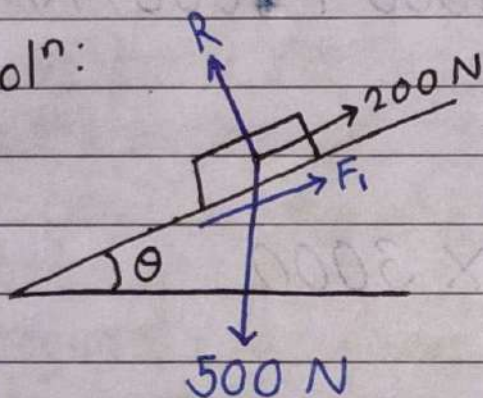
For impending motion of B,

$$P = F_1 + F_2 = 250 + 1000$$

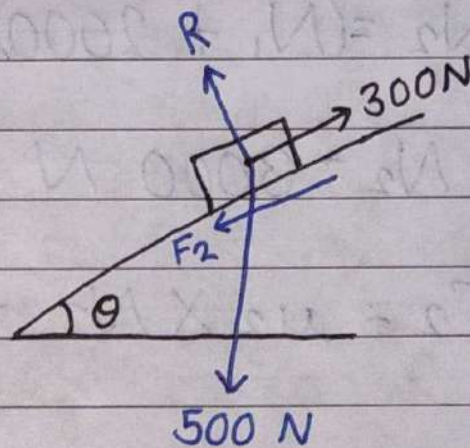
$$\therefore P = 1250 \text{ N.}$$

9) A block of weight 500 N just starts moving down a rough inclined plane when it is subjected to 200 N force, acting up the inclined plane, and it is at the point of moving up the plane when ~~could~~ it is pulled up by a force of 300 N parallel to the plane. Find the inclination of the plane and the coefficient of friction between the inclined plane and the block.

Solⁿ:



Case - I



Case - II

For both cases,

$$R = 500 \cos \theta$$

Also,

$$F = \mu R = \mu(500 \cos \theta)$$

Case - I:

$$200 + F = 500 \sin \theta$$

$$\Rightarrow 200 + \mu(500 \cos \theta) = 500 \sin \theta$$

$$\Rightarrow \mu(500 \cos \theta) = 500 \sin \theta - 200 \quad \text{--- (1)}$$

Case - II:

$$300 = F + 500 \sin \theta$$

$$\Rightarrow 300 = \mu (500 \cos \theta) + 500 \sin \theta \quad \text{---(2)}$$

From eqⁿ. (1), eqⁿ. (2) becomes,

$$300 = 500 \sin \theta - 200 + 500 \sin \theta$$

$$\Rightarrow 300 + 200 = 1000 \sin \theta$$

$$\Rightarrow 500 = 1000 \sin \theta$$

$$\Rightarrow \sin \theta = \frac{500}{1000}$$

$$\Rightarrow \sin \theta = 0.5$$

$$\Rightarrow \theta = \sin^{-1}(0.5)$$

$$\therefore \theta = 30^\circ$$

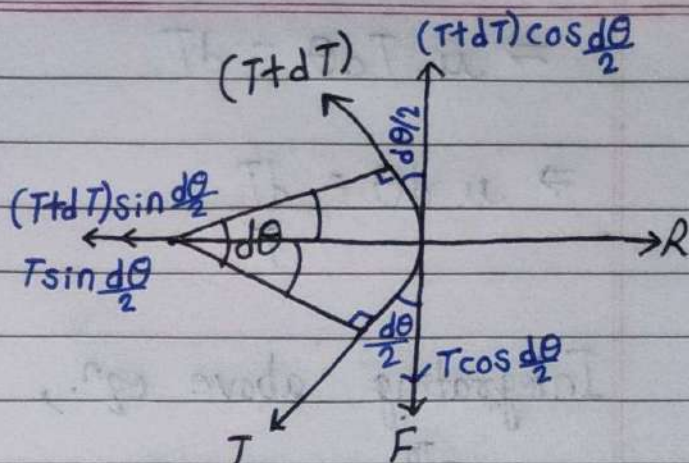
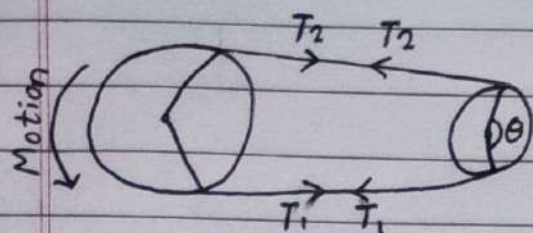
From eqⁿ. (1),

$$\mu (500 \cos 30) = 500 \sin 30 - 200$$

$$\Rightarrow \mu = \frac{500 \sin 30 - 200}{500 \cos 30}$$

$$\therefore \mu = 0.11547$$

Imp. # Belt Friction



From figure,

$$R = (T + dT) \sin \frac{d\theta}{2} + T \sin \frac{d\theta}{2}$$

$$\Rightarrow R = T \cdot \sin \frac{d\theta}{2} + dT \cdot \sin \frac{d\theta}{2} + T \sin \frac{d\theta}{2}$$

$$\Rightarrow R = T \cdot \frac{d\theta}{2} + T \cdot \frac{d\theta}{2} \quad \left[\because dT \text{ \& } d\theta \rightarrow \text{very small} \right]$$

$$\Rightarrow \sin \frac{d\theta}{2} \rightarrow \frac{d\theta}{2}$$

$$\Rightarrow R = 2T \cdot \frac{d\theta}{2}$$

$$\Rightarrow R = T \cdot d\theta$$

Also, from figure,

$$(T + dT) \cos \frac{d\theta}{2} - F - T \cos \frac{d\theta}{2} = 0$$

$$\Rightarrow \cancel{T \cos \frac{d\theta}{2}} + dT \cos \frac{d\theta}{2} - F - \cancel{T \cos \frac{d\theta}{2}} = 0$$

$$\Rightarrow dT \times 1 - F = 0 \quad \left[\because \frac{d\theta}{2} \rightarrow \text{very small, } \cos \frac{d\theta}{2} = 1 \right]$$

$$\Rightarrow F = dT$$

$$\Rightarrow \mu \cdot R = dT \quad \left[\because \mu = F/R \right]$$

$$\Rightarrow \mu \cdot T d\theta = dT$$

$$\Rightarrow \mu d\theta = \frac{dT}{T}$$

Integrating above eqn.,

$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^\theta \mu d\theta$$

$$\Rightarrow [\log T]_{T_1}^{T_2} = \mu [\theta]_0^\theta$$

$$\Rightarrow \log T_2 - \log T_1 = \mu (\theta - 0)$$

$$\Rightarrow \log \frac{T_2}{T_1} = \mu \theta$$

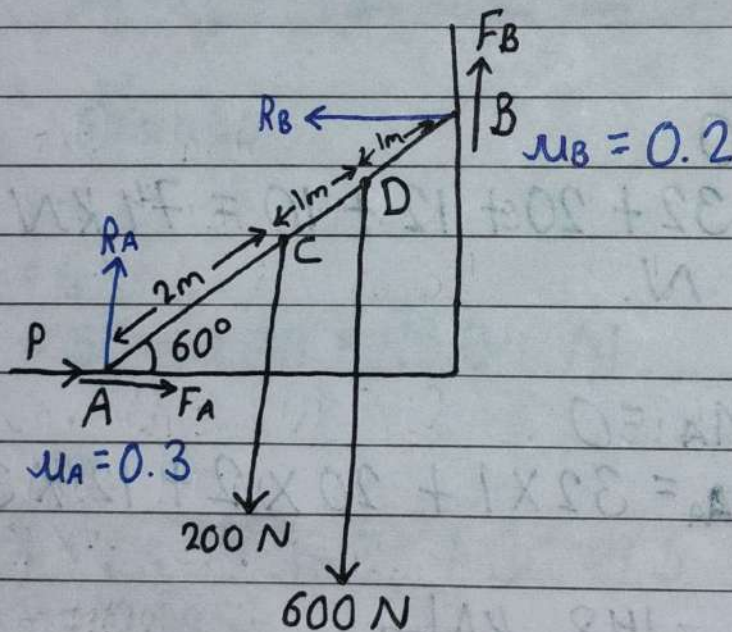
$$\Rightarrow \boxed{\frac{T_2}{T_1} = e^{\mu \theta}}$$

This is the required equation.

Numericals from Ladder Friction

- Q) A ladder of length 4m weighing 200 N is placed against a vertical wall as shown in figure. The coefficient of friction between the wall and the ladder is 0.2 and that between floor and the ladder is 0.3. The ladder in addition to its own weight has to support a man weighing 600 N at a distance of 3m from A. Calculate the maximum horizontal force to be applied at A to prevent slipping?

Solⁿ:



$$F_A = \mu_A R_A \quad \& \quad F_B = \mu_B R_B$$

$$\sum M_A = 0$$

$$\Rightarrow 200 \cdot \cos 60^\circ \times 2 + 600 \cdot \cos 60^\circ \times 3 - R_B \cdot \sin 60^\circ \times 4 - F_B \cdot \cos 60^\circ \times 4 = 0$$

$$\Rightarrow 1100 = R_B \times 3.4641 + R_B \times 0.4$$

$$\Rightarrow R_B = \frac{1100}{3.8641} = 284.67 \text{ N}$$

$$\sum V = 0$$

$$\Rightarrow R_A + F_B - 200 - 600 = 0$$

$$\Rightarrow R_A + 0.2 \times R_B - 800 = 0$$

$$\Rightarrow R_A = 800 - 0.2 \times 284.67$$

$$\Rightarrow R_A = 743.066 \text{ N}$$

$$\sum H = 0$$

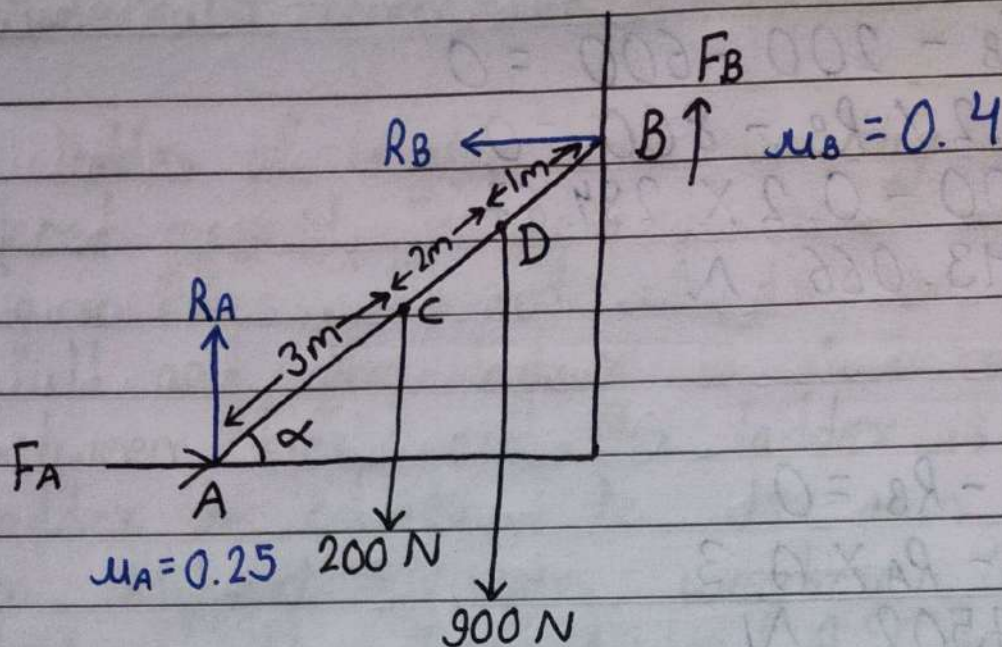
$$\Rightarrow P + F_A - R_B = 0$$

$$\Rightarrow P = R_B - R_A \times 0.3$$

$$\Rightarrow P = 61.7502 \text{ N}$$

Hence, the maximum horizontal force required at A to prevent slipping is 61.75 N.

- Q) The ladder shown in figure below is 6m long and is supported by a horizontal floor and a vertical wall. The coefficient of friction between the floor and the ladder is 0.25 and between the wall and the ladder is 0.4. The weight of the ladder is 200 N and may be considered as a concentrated load at G. The ladder supports a vertical load of 900 N at C which is at a distance of 1m from B. Determine the least value of α at which the ladder may be placed without slipping. Determine the reaction at that stage.



$$F_A = \mu_A R_A = 0.25 R_A$$

$$F_B = \mu_B R_B = 0.4 R_B$$

$$\sum H = 0$$

$$\Rightarrow F_A - R_B = 0$$

$$\Rightarrow F_A = R_B \Rightarrow R_B = 0.25 R_A$$

$$\sum V = 0$$

$$\Rightarrow R_A + F_B - 200 - 900 = 0$$

$$\Rightarrow R_A + 0.4 R_B - 1100 = 0$$

$$\Rightarrow R_A + 0.4 \times 0.25 R_A = 1100$$

$$\Rightarrow 1.1 R_A = 1100$$

$$\Rightarrow R_A = 1000 \text{ N}$$

$$R_B = 0.25 \times 1000 = 250 \text{ N}$$

$$F_A = 0.25 \times 1000 = 250 \text{ N}$$

$$F_B = 0.4 \times 250 = 100 \text{ N}$$

$$\sum M_A = 0$$

$$\Rightarrow (200 \times 3 \cos \alpha) + (900 \times 5 \cos \alpha) - (R_B \times 6 \sin \alpha) - (F_B \times 6 \cos \alpha) = 0$$

$$\Rightarrow 600 \cos \alpha + 4500 \cos \alpha - 1500 \sin \alpha - 600 \cos \alpha = 0$$

$$\Rightarrow 4500 \cos \alpha = 1500 \sin \alpha$$

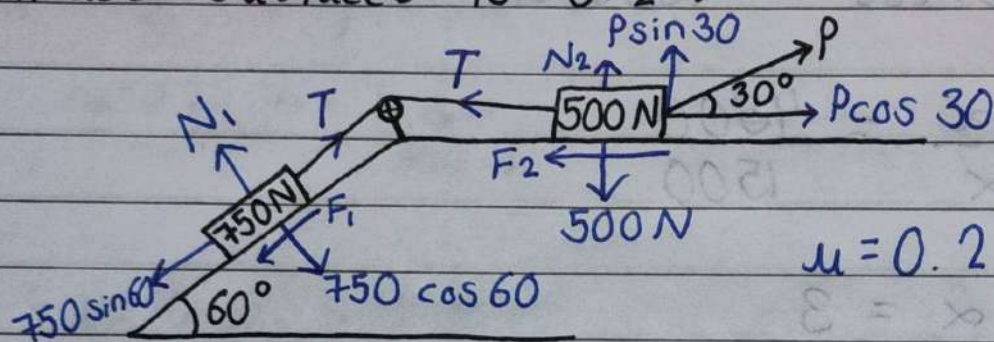
$$\Rightarrow \frac{\sin \alpha}{\cos \alpha} = \frac{4500}{1500}$$

$$\Rightarrow \tan \alpha = 3$$

$$\therefore \alpha = 71.565^\circ$$

Numericals from Friction

Imp. Q) What is the value of P in the system as shown in figure to cause the motion to impend? Assume the pulley is smooth and the coefficient of friction between the other contact surfaces is 0.2.



Solⁿ:

For 750 N Block,

$$N_1 = 750 \cos 60 = 375 \text{ N}$$

$$F_1 = \mu N_1 = 0.2 \times 375 = 75 \text{ N}$$

$$T = 750 \sin 60 + F_1 = 724.519 \text{ N}$$

For 500 N Block,

$$N_2 + P \sin 30 = 500$$

$$\Rightarrow N_2 = 500 - P \times \frac{1}{2}$$

$$F_2 = \mu N_2 = 0.2 \left(500 - \frac{P}{2} \right)$$

$$\Rightarrow F_2 = 100 - 0.1P$$

$$T + F_2 = P \cos 30$$

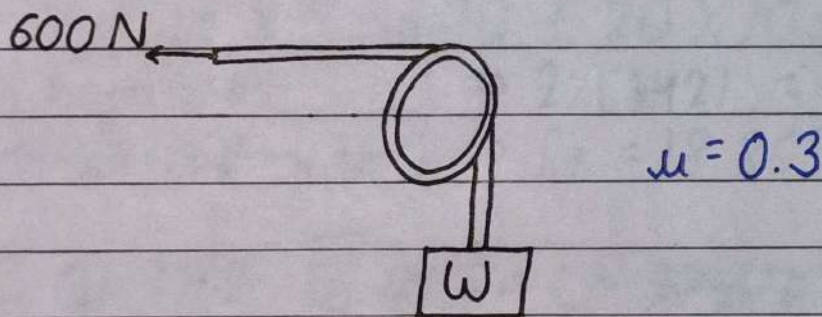
$$\Rightarrow 724.519 + 100 - 0.1 \times P = P \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow 824.519 = 0.9660 \times P$$

$$\Rightarrow P = 853.5393 \text{ N.}$$

Q) A rope making $1\frac{1}{4}$ turns around a stationary horizontal drum is used to support a weight W . If the $\mu = 0.3$, what range of weight can be supported by exerting a 600 N force at the other end of the rope?

Solⁿ:



(i) Let the motion be ~~in~~ downward,

$$\text{Angle of contact} = \frac{5}{4} \times 2\pi = 2.5\pi$$

We have,

$$\frac{W}{600} = e^{0.3 \times 2.5\pi}$$

$$\left[\text{From Belt Equation, } \frac{T_2}{T_1} = e^{\mu\theta} \right]$$

$[T_2 > T_1]$

$$\Rightarrow W = 6,330.434 \text{ N}$$

(ii) Let the motion be upward,

$$\frac{600}{W} = e^{0.3 \times 2.5 \pi}$$

$$\Rightarrow W = 56.868 \text{ N.}$$

Hence the range of weight is 56.868 N to 6,330.434 N.

